

2520090088(B.DHARSHAN)

SKILL 10

```
root@PC: ~/os_lab/pwd_exit_export_lab
cd ~/os_lab
mkdir -p pwd_exit_export_lab && cd pwd_exit_export_lab
root@PC:~# cd ~/os_lab
root@PC:~/os_lab# mkdir -p pwd_exit_export_lab && cd pwd_exit_export_lab
root@PC:~/os_lab/pwd_exit_export_lab# cd ~/os_lab
mkdir -p pwd_exit_export_lab && cd pwd_exit_export_lab
root@PC:~/os_lab/pwd_exit_export_lab# nano pwd_exit.c
root@PC:~/os_lab/pwd_exit_export_lab# gcc -g -O0 -Wall -pthread pwd_exit.c -o pwd_exit
./pwd_exit
cat shell_state.txt
=== Task 1: PWD and Exit State Handling ===

[EXEC] Running 'pwd':
/root/os_lab/pwd_exit_export_lab

[EXEC] Simulating user command 'exit 0':
[EXIT] Processing exit request (Exit Code: 0)...
[STATE] Session state saved to 'shell_state.txt'.
[CLEANUP] Allocated history buffer successfully freed.
[EXIT] Exiting shell process now.
LAST_CWD=/root/os_lab/pwd_exit_export_lab
STATUS=CLEAN_EXIT
root@PC:~/os_lab/pwd_exit_export_lab# nano export_handler.c
root@PC:~/os_lab/pwd_exit_export_lab# gcc -g -O0 -Wall -pthread export_handler.c -o
export_handler
./export_handler
=== Task 2: Export Syntax Parsing and Inheritance Test ===

[TEST 1] Exporting valid variable 'LAB_USER=engineering_student':
[EXPORT SUCCESS] Updated environment: LAB_USER=engineering_student

[TEST 2] Exporting valid variable without value 'EMPTY_VAR':
[EXPORT SUCCESS] Updated environment: EMPTY_VAR=

[TEST 3] Exporting invalid identifier '123_INVALID=test':
export: '123_INVALID': not a valid identifier

[TEST 4] Exporting invalid identifier 'BAD-NAME=test':
export: 'BAD-NAME': not a valid identifier
```

root@PC: ~/os_lab/pwd_exit_export_lab

```
[TEST 3] Exporting invalid identifier '123_INVALID=test':  
export: '123_INVALID': not a valid identifier
```

```
[TEST 4] Exporting invalid identifier 'BAD-NAME=test':  
export: 'BAD-NAME': not a valid identifier
```

```
[TEST 5] Testing child process inheritance via fork() + execvp():
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```
[CHILD PID 574] Querying inherited environment for 'LAB_USER':
```

```
engineering_student
```

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root@PC:~/os_lab/pwd_exit_export_lab# valgrind --leak-check=full ./pwd_exit
```

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valgrind --leak-check=full ./export_handler
```

```
==577== Memcheck, a memory error detector
```

```
==577== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
```

```
==577== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
```

```
==577== Command: ./pwd_exit
```

```
==577==
```

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=== Task 1: PWD and Exit State Handling ===
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```
[EXEC] Running 'pwd':
```

```
/root/os_lab/pwd_exit_export_lab
```

```
[EXEC] Simulating user command 'exit 0':
```

```
[EXIT] Processing exit request (Exit Code: 0)...
```

```
[STATE] Session state saved to 'shell_state.txt'.
```

```
[CLEANUP] Allocated history buffer successfully freed.
```

```
[EXIT] Exiting shell process now.
```

```
==577==
```

```
==577== HEAP SUMMARY:
```

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==577==    in use at exit: 0 bytes in 0 blocks
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```
==577==   total heap usage: 4 allocs, 4 frees, 5,620 bytes allocated
```

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==577==
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```
==577== All heap blocks were freed -- no leaks are possible
```

```
==577==
```

```
==577== For lists of detected and suppressed errors, rerun with: -s
```

```
==577== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```
==578== Memcheck, a memory error detector
```

```
==578== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
```

```
==578== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
```

```
==578== Command: ./export_handler
```

```
==578==
```

```
root@PC: ~/os_lab/pwd_exit_export_lab

==577== For lists of detected and suppressed errors, rerun with: -s
==577== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
==578== Memcheck, a memory error detector
==578== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==578== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==578== Command: ./export_handler
==578==

=== Task 2: Export Syntax Parsing and Inheritance Test ===

[TEST 1] Exporting valid variable 'LAB_USER=engineering_student':
[EXPORT SUCCESS] Updated environment: LAB_USER=engineering_student

[TEST 2] Exporting valid variable without value 'EMPTY_VAR':
[EXPORT SUCCESS] Updated environment: EMPTY_VAR=

[TEST 3] Exporting invalid identifier '123_INVALID=test':
export: '123_INVALID': not a valid identifier

[TEST 4] Exporting invalid identifier 'BAD-NAME=test':
export: 'BAD-NAME': not a valid identifier

[TEST 5] Testing child process inheritance via fork() + execvp():
[CHILD PID 579] Querying inherited environment for 'LAB_USER':
engineering_student
==578==
==578== HEAP SUMMARY:
==578==   in use at exit: 0 bytes in 0 blocks
==578== total heap usage: 11 allocs, 11 frees, 1,718 bytes allocated
==578==
==578== All heap blocks were freed -- no leaks are possible
==578==
==578== For lists of detected and suppressed errors, rerun with: -s
==578== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@PC:~/os_lab/pwd_exit_export_lab# # Export in parent subshell and modify in child subshell
export TEST_VAR=PARENT && (export TEST_VAR=CHILD) && echo "Parent Value: $TEST_VAR"
Parent Value: PARENT
root@PC:~/os_lab/pwd_exit_export_lab#
```